

SAMUEL AMOAH

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EDUCATION

University of Ghana

Legon, Ghana

BSc. Agricultural Engineering | Second Class Upper (Hons) GPA: 3.54/4

Graduated: January 2024

Relevant Coursework: Soil Mechanics, Fluid Mechanics, Irrigation & Drainage Engineering, Hydrology and Water Resources Engineering, Statistics for Engineers, Calculus 1, Numerical Methods, Technical Reports Writing

TECHNICAL SKILLS

Data Analytics Tools: SQL, Tableau, Microsoft Excel (advanced), Power BI (basic), SPSS

Programming & Data: Python (pandas, NumPy, matplotlib, Jupyter Notebook), R (statistical analysis), JavaScript, HTML/CSS

Data Analysis Process: Data cleaning, data transformation, exploratory data analysis, data visualization, dashboard design, statistical analysis

Software & Platforms: AquaCrop modelling software, SolidWorks, VS Code, Git & GitHub, Microsoft Office Suite (Word, PowerPoint, Excel)

Soft Skills: Data interpretation, research & analysis, attention to detail, communication, teamwork, adaptability

PROJECT EXPERIENCE

Cyclistic Bike-Share Case Study — Google Data Analytics Capstone

2026

- Analyzed over 5.6 million ride records to uncover behavioral differences between member and casual riders, using Python and Jupyter Notebook to clean, transform, and merge twelve months of raw trip data.
- Identified that members took 3,668,146 rides versus 2,013,334 for casual riders, while casual riders averaged longer trips (19.6 minutes vs. 12.3 minutes), pointing to leisure-driven usage patterns.
- Built an interactive Tableau dashboard visualizing ride volume, duration, hourly and weekly trends, and popular station locations to support data-driven marketing recommendations.
- Surfaced a casual-rider peak between 4–8 PM and weekend afternoons, informing a proposed strategy to convert casual riders into annual members through targeted weekend promotions.

AquaCrop Crop Yield Modelling — BSc Capstone

University of Ghana | May – Jul 2023

- Compared actual vs. predicted lettuce yields using AquaCrop; achieved RMSE of 0.09 (actual: 13.46 t/ha, predicted: 13.55 t/ha) — demonstrating strong quantitative modelling accuracy.
- Analysed climatic, soil, and irrigation data to simulate crop growth scenarios, informing sustainable water management strategies under water-scarcity conditions.
- Applied wastewater treated with corn cob biochar filtration for irrigation — validating eco-efficient agricultural data models.

WORK EXPERIENCE

Teaching Assistant – Soil Mechanics & Fluid Mechanics

Nov 2023 – Nov 2024

University of Ghana, Legon, Ghana

- Delivered structured learning sessions to 150+ students, boosting participation by 30% through data-driven teaching methods.
- Co-developed course assessment materials that raised average student scores by 15% semester-on-semester.
- Streamlined grading workflows, reducing turnaround time by 40% and improving student satisfaction ratings by 25%.
- Invalidated examinations for 200+ students during the 2024 School of Engineering Sciences examination period.

Engineering Intern

Sep 2022 – Nov 2022

Modern Engineering Power Solutions Ltd (MEPS), Pokuase Rd., Ghana

- Inspected 5+ generators and flagged wear/damage patterns, contributing to a 15% reduction in unexpected downtime.
- Executed routine maintenance on 8+ generator units, achieving a 10% improvement in operational efficiency.
- Collaborated on diagnosing mechanical faults, enabling critical repairs on three high-capacity generators within 48 hours.

COMMUNITY SERVICE & EXTRACURRICULAR ACTIVITIES

STEM Education Volunteer

2024

Annual Science-in-Tents Event, Noguchi

- Promoted STEM education and career pathways to 100 Junior High School students, delivering engaging sessions on opportunities in engineering, data science, and technology.

CERTIFICATIONS

- Google Data Analytics Professional Certificate — Coursera / Google Career Certificates (June 2026)
- Web Development Fundamentals — IBM SkillsBuild (March 2025)
- AI Career Essentials — ALX Africa (July 2024)
- Ghana National Service Certificate (October 2024)

LANGUAGES

English: Native | French: Beginner